

Application of Self Organizing Map for Intelligent Machine Fault Diagnostics Based on Infrared Thermography Images

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Abstract—This paper concerns with implementation of self organizing map (SOM) for intelligent machine fault diagnostics. The present study employs infrared images acquired by thermography camera as data base of machine diagnostics system. Image processing is carried out using thresholding for image segmentation and clustering by means of *k*-means algorithm. Feature extraction of images is conducted by calculating area, perimeter and central moment of region of interest (ROI). All data of this work was acquired by capturing the images of rolling element bearings from rotating machine fault simulator (MFS). The simulator is able to experiment a normal and seeded fault conditions such as outer and inner race defects of rolling element bearing, unbalance, misalignment and looseness. Pattern recognition technique is then employed to diagnose the machine conditions by mapping the image features through SOM. The result shows that SOM based infrared thermography image can perform intelligent machine fault diagnostics with plausible accuracy.

Keywords—self organizing map; machine fault diagnostics; infrared thermography; pattern recognition

I. INTRODUCTION

The most machinery used in industry is rotating machines such as turbine, pump, compressor, blower and motor. Due to increasing massive production in industry, it is very meaningful to uninterrupted operation of machines during their productivity. Unwanted downtime due to machine failure leads loss of product and profit for industry. Therefore, machine condition monitoring and fault diagnostics become serious issue in maintenance activity to serve the demand of availability and avoid operator casualties as well as economic losses [1]. Several sensors has been used such as accelerometer, eddy current probe, acoustic emission, thermocouple and infrared thermography (IRT) to fulfill the requirement of machine condition monitoring [2]. The assessment of critical components in the machines like bearing (journal or rolling element) using such sensors is important to early detection of machine condition.

Among condition monitoring tools, IRT is popular due to simple and easy in use and interpretation, non-contact, and free from dangerous radiation [3]. Moreover, the application of IRT for condition monitoring has covered wide area such as petrochemical, electrical, mechanical, environmental and even medical application [3-7]. In mechanical engineering, IRT allows diagnostics of machineries through thermal patterns of pumps, motors,

bearings, fans, blowers, pulley, shaft and other elements [8]. Thermal pattern is able to indicate whether the machines condition is normal or abnormal. For example, the bearing support contains very useful information subject to machine condition. Bearing is still the best location for data acquisition that represents the machine state. Therefore, machine condition monitoring in real application should be experimented on them. IRT image approach has been reported for non-destructive (NDT) evaluation for mechanical system, e.g., to check misalignment, bent shaft and rolling element bearing [9]. Hence, data analysis technique is essential for all methods of machine condition identification.

This paper deals with implementation of intelligent system through SOM for machine diagnostics based on infrared thermography image. Image processing is performed by thresholding for image segmentation and clustering by means of *k*-means algorithm. Salient image features are extracted by calculating the area, perimeter and central moment of ROI (hottest region).

II. BASIC THEORY

A. Thermal Image

Infrared camera does not measure directly a temperature of surface. The detectors of camera are sensitive to the luminance emitted by the object being studied. This luminance is naturally dependent on the temperature of surface. Thermal image is an image which is captured by infrared thermal camera. This device can capture electromagnetic spectrum within infrared bands, 0.78 – 1000 μm . A thermal image is a function of radiated energy on an inspected object [10].

B. Image Thresholding

Image thresholding is mostly used for image segmentation [11]. Thresholding is also common method for defect detection of material by IRT image. Thresholding is a process to separate objects from its background in a digital image. Histogram is the main tool in this separation process. Suppose the grey level corresponds to an image $f(x,y)$ that composed of lights objects on a dark background, in such a way that object and background pixels have grey levels grouped into two dominant modes. Extraction the object from the background is performed by selecting a threshold T that

separates these modes. A thresholded image $g(x,y)$ is defined as

$$g(x,y) = \begin{cases} 1 & \text{if } f(x,y) > T \\ 0 & \text{if } f(x,y) \leq T \end{cases} \quad (1)$$

The method of global threshold is selected in this paper where the grey-level histogram is calculated from the entire image [11].

C. Image Feature Extraction

- Perimeter and area are meaningful for binary image. Consider a discrete binary image containing one more objects, where $f(j,k) = 1$ if a pixel is part of the object and $f(j,k) = 0$ for all non-object or background pixels. The Bit-Quads [12] method is adopted to determine the perimeter and area of ROI.
- The moment of order $(p+q)$ of an image [13], denoted as m_{pq} , is defined as

$$m_{pq} = \sum_x \sum_y x^p y^q g(x,y) \quad (2)$$

where x and y are coordinate of the pixel, and $g(x,y)$ is the grey level of the pixel. The higher order moments represent spatial distribution of the objects. The above two characteristics make moments a reasonable, proper feature for the classification task. However, the above equation may be greatly affected by objects shifting. Therefore, to overcome this problem central moment is used and defined as

$$\mu_{pq} = \sum_x \sum_y (x - \bar{x})^p (y - \bar{y})^q g(x,y) \quad (3)$$

where $\bar{x} = m_{10} / m_{00}$ and $\bar{y} = m_{01} / m_{00}$.

The relationship between moments and central moments up to order 3 are listed as follows

$$\begin{aligned} \mu_{00} &= m_{00} & \mu_{11} &= m_{11} - \bar{y}m_{10} \\ \mu_{10} &= 0 & \mu_{30} &= m_{30} - 3\bar{x}m_{20} + 2\bar{x}^2m_{10} \\ \mu_{01} &= 0 & \mu_{12} &= m_{12} - 2\bar{y}m_{11} - \bar{x}m_{02} + 2\bar{y}^2m_{10} \\ \mu_{20} &= m_{20} - \bar{x}m_{10} & \mu_{21} &= m_{21} - 2\bar{x}m_{11} - \bar{y}m_{02} + 2\bar{x}^2m_{01} \\ \mu_{02} &= m_{02} - \bar{y}m_{01} & \mu_{03} &= m_{03} - 3\bar{y}m_{02} + 2\bar{y}^2m_{01} \end{aligned} \quad (4)$$

- The SOFM is a neural network model that implements a characteristic non-linear projection from a high dimensional space of sensory or other input signals onto a low-dimensional array of neurons. The SOFM is an unsupervised learning algorithm which produces a map pattern features on its output layer [14]. Input patterns with similar features are mapped onto neighbouring output nodes. The network consists of an input layer with m neurons and an output layer with n neurons as shown in Fig. 1.

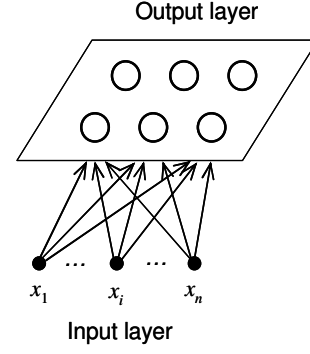


Figure 1. The SOM network.

The image of the signal space tends to manifest the clusters of input information and their relationship on the feature map. With every node j , a parametric reference vector $\mathbf{w}_j \in R^n$ is associated. In an abstract scheme it may be imagined that at time k the input $\mathbf{x}(k)$, by means of some parallel computing mechanisms, is compared with all $\mathbf{w}_j(k)$, and the location of best match in some metric is defined as the location of the response. However, in many practical applications, the Euclidean distances $\|\mathbf{x}(k) - \mathbf{w}_j(k)\|$ can be made to define the best-matching node (neuron) i which is signified by the index $i(\mathbf{x})$

$$\begin{aligned} i(\mathbf{x}) &= \arg \min \|\mathbf{x}(k) - \mathbf{w}_j(k)\| \\ &= \arg \min \{\|\mathbf{e}_j(k)\|\} \end{aligned} \quad (5)$$

where $j = 1, 2, \dots, N$; $k = 1, 2, \dots, M$; $\mathbf{x}(k) = \{x_1, x_2, \dots, x_M\}^T$ represents the input vectors and $\mathbf{w}_j(k) = [w_{j1}, w_{j2}, \dots, w_{jM}]^T$ ($j = 1, 2, \dots, N$) represents the weight vectors of the j th output vectors. $\mathbf{e}_j(k)$ is the error which is given by $\|\mathbf{x}(k) - \mathbf{w}_{i(\mathbf{x})}(k)\| = \min\{\|\mathbf{e}_j(k)\|\}$. The detail of SOM theory was reported in [14].

III. METHOD AND MATERIAL

The flowchart of diagnostics process is presented in Fig. 2. After image acquisition from object, thermal image is cropped to select the ROI that is rolling element bearing image. Next step, image clustering will be performed by using k -means algorithm to cluster the hotspot area based on color palette. Image segmentation by thresholding is conducted using Otsu method [15]. Image thresholding will produce binary image which useful for image feature extraction. Perimeter, area and central moments are then calculated based on resulted binary images. After feature extraction, extracted features from normal and faulty condition are separated for training SOM purpose. The feature of normal condition will be used as baseline of diagnostics process. Finally, SOM will make decision of machine condition normal or fault.

Data acquisition was conducted on MFS which is able to simulate abnormality in rotating machine i.e., rolling element bearing fault, unbalance, misalignment, and looseness. Thermography images were captured from the machine that has been operated for 30 minutes to generate

initial heat in the bearings. FLIR SC5000 infrared camera was used to capture the images of bearing with eight simulation conditions. The acquisition produced 30 images per second and machine was operated at 1800 rpm. Figs. 3 and 4 show the data acquisition process in the laboratory and thermal image of MFS, respectively.

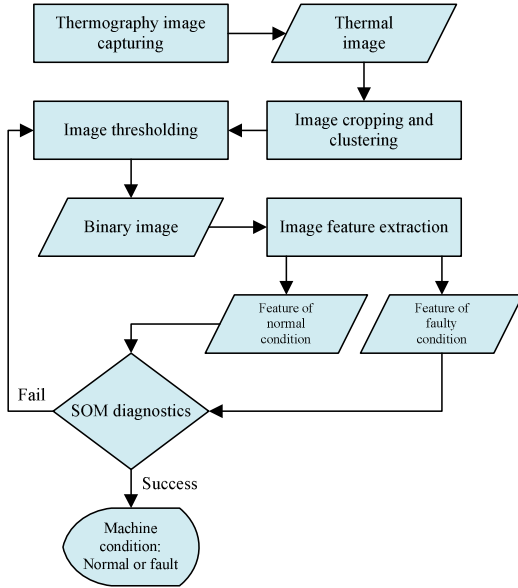


Figure 2. Flow chart of SOM diagnostics process.



Figure 3. Data acquisition process.

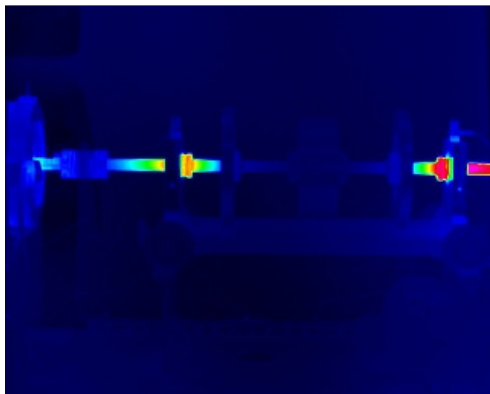


Figure 4. Thermal images of MFS.

IV. RESULT AND DISCUSSION

After cropping, thermal images captured from right and left bearings are presented in Figs. 5 and 6. The ROI is determined by red or pink colors that represent hottest temperature area generated by bearings. Such colors were then extracted and clustered by k -means algorithm with $k = 6$. Fig. 7 depicts the clustering result by means of k -means algorithm.

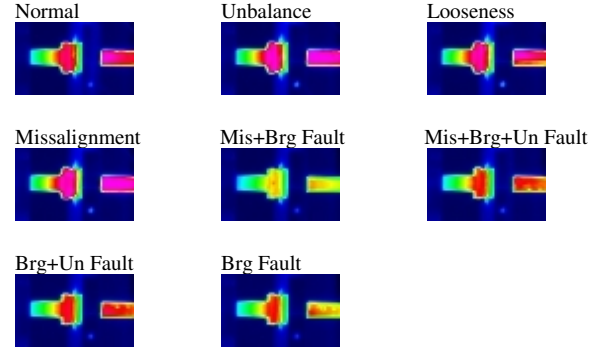


Figure 5. Thermal images of right-bearing.

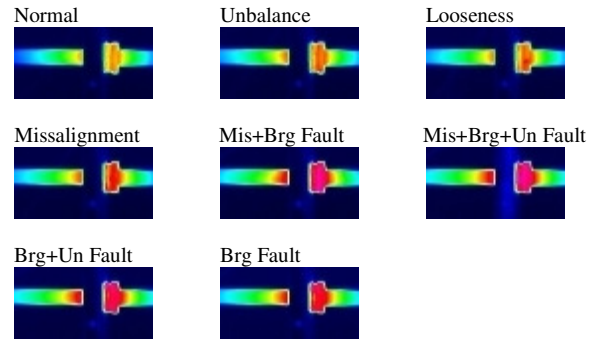


Figure 6. Thermal images of left-bearing.

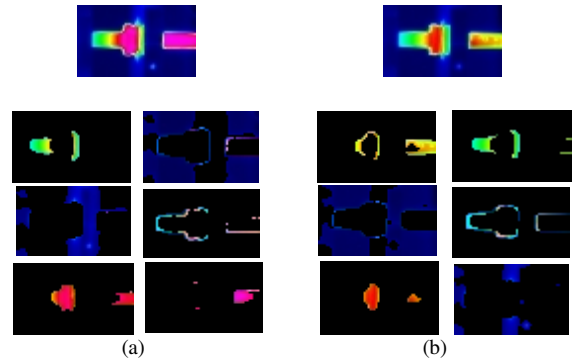


Figure 7. Clustering of bearing images: a. Normal bearing; b. Faulty bearing

Clustering image based on colors produced ROI from which feature calculation will be performed. The selected ROI will be transformed into binary image by thresholding method. Next step is feature calculation of perimeter, area and central moments of ROI. This process is shown in Fig. 8.

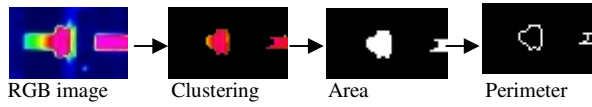


Figure 8. Processing of bearing thermal image.

The ROI area is then plotted based on pixel and data number to know the patterns of each machine condition. Figs. 8 and 9 show the patterns of area calculated from right and left bearings, respectively.

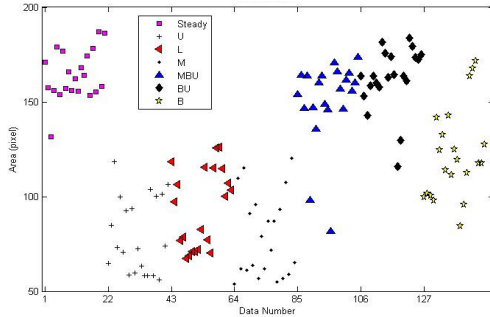


Figure 8. Feature area of right-bearing.

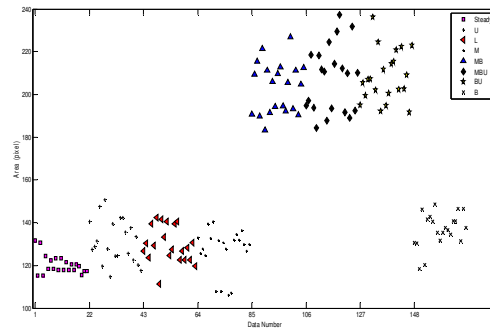


Figure 9. Feature area of left-bearing.

Both figure represents patterns for data input of classification algorithm such as neural network for machine diagnostics purpose. Observing these figures, the data shows good separation of machine condition. However, some conditions such as bearing fault and combination fault of misalignment, bearing fault and unbalance are overlap. In addition, plot of perimeter is presented in Figs. 10 and 11.

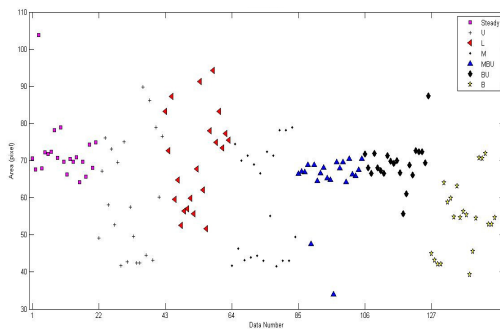


Figure 10. Feature perimeter of right-bearing.

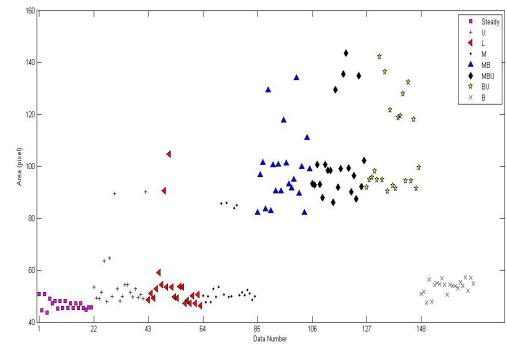


Figure 11. Feature perimeter of left-bearing.

Fault diagnostics process is started by training the SOM using data input obtained from image features. The presentation of unified-matrix (U-matrix) after training for right bearing is depicted in Fig. 12a. Fig. 12b shows trained SOM for feature area.

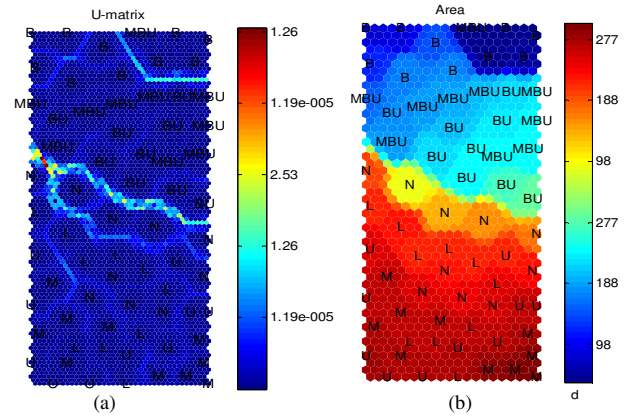


Figure 12. SOM of right-bearing: a. U-matrix; b. Trained SOM for area

From the U-matrix, it easy to see that top row is dominated bearing fault then followed by combination fault of misalignment, bearing fault and unbalance. The normal condition is located at the center then looseness and the bottom of U-matrix is dominated by misalignment condition. The similar trained SOM is also shown by perimeter and m_{00} feature as depicted in Fig. 13. It also shows that the map of feature area is similar with m_{00} .

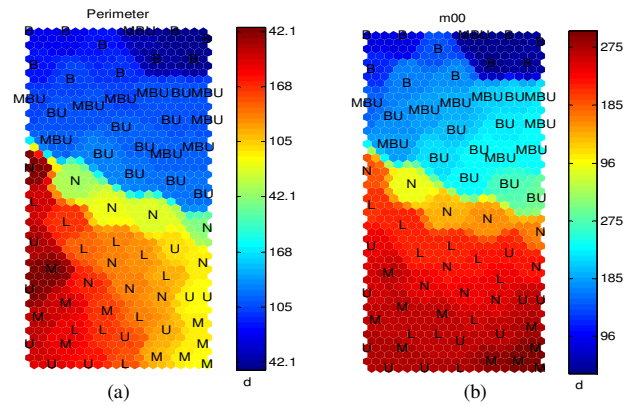


Figure 13. Trained SOM: a. Perimeter; b. m_{00} .

After training, several features are set to test the trained SOM. As representation of the result, Figs. 14 and 15 show the testing of SOM based on area, perimeter and m_{00} features. Feature area shows good result in testing with matching conditions similar with trained SOM. Perimeter and m_{00} features show worse performance than feature area due to lack of matching conditions. It represents that feature area is salient feature for image and is more reliable than perimeter and m_{00} .

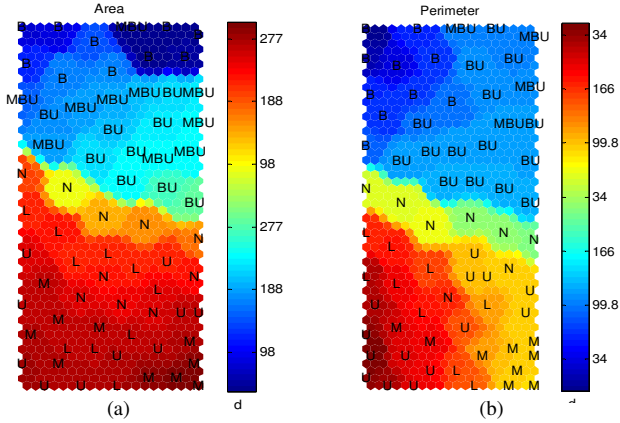


Figure 14. Tested SOM: a. Area; b. Perimeter

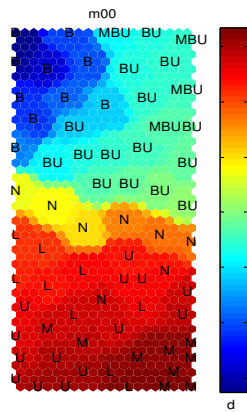


Figure 15. Tested SOM: m_{00} feature.

Table I summarizes the accuracies of diagnostics process based on extracted image features for right.

TABLE I. DIAGNOSTICS ACCURACY OF RIGHT-BEARING

Machine condition	Accuracy (%)		
	Area	Perimeter	m_{00}
Normal (N)	70	80	100
Unbalance (U)	n/a	n/a	100
Loseness (L)	n/a	40	n/a
Misalignment (M)	100	n/a	n/a
Bearing fault (B)	40	70	40
M+B	n/a	n/a	n/a
M+B+U	70	n/a	20
B+U	30	n/a	80

SOM can successfully diagnoses the normal, misalignment and combination fault of misalignment, bearing fault and unbalance by using feature area. Normal and bearing fault condition can be detected by SOM through feature area, perimeter and m_{00} . Some diagnostics process are failed to correctly diagnose the machine conditions. It is shown by 'n/a' mark in the presented table. Low accuracy in this experiment is caused by the size of ROI is too small compared with the background resulted in the data acquisition. The other reason is the machine conditions (8 conditions) are redundant for diagnostics process.

V. CONCLUSIONS

This study concludes that machine diagnostics based on infrared thermography can be performed using intelligent system such as SOM algorithm. Image feature extraction is needed to obtain salient feature e.g., area, perimeter and central moment of ROI. The ROI size should be considered to produce clear thermal image for good image feature extraction. Automatic machine fault diagnostics is difficult to be realized when the machine conditions being diagnosed is too complex.

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